

A constructive ATLAS of finite simple groups in Lean

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Abstract

We present a constructive atlas of finite simple groups with proofs of their orders, simplicity, and structural properties. The eight completed families are cyclic groups of prime order, alternating groups, the classical series $A_r(q)$, $B_r(q)$, $C_r(q)$, $D_r(q)$, and the exceptional series $G_2(q)$ and the small Ree groups ${}^2G_2(3^{2m+1})$, $m \geq 1$. The fifteen sporadic entries are M_{11} , M_{12} , M_{22} , M_{23} , M_{24} , Co_1 , Co_2 , Co_3 , McL , HS , Suz , J_2 , and Fi_{22} , Fi_{23} , Fi'_{24} . The simple parameter ranges and exceptional cases are stated explicitly. The models arise from codes, lattices, forms, algebras, and finite geometries, including the split octonions and the Conway–Parker algebra. They retain natural actions, stabilizers, central quotients, and comparison maps for subsequent group theory. Classical comparison isomorphisms relate models, while involution-class counts distinguish the equal-order orthogonal and symplectic families in odd characteristic and rank at least three. Drawing on classical sources and companion mathematical work, the project develops the construction side of finite simple group theory, not the exhaustiveness proof of the classification. The models and proofs are formalized in Lean for verification and reuse.

Source repository: <https://github.com/Moonshine-in-Kansas/constructive-atlas>

1 Introduction

The construction side of finite simple group theory asks for concrete models together with proofs of their orders, simplicity, and characteristic structure. Codes, lattices, algebras, forms, and finite geometries provide such models and retain natural actions, stabilizers, central quotients, and maps between related objects. The ATLAS project organizes these constructions into a structural library, with proofs formalized in Lean [28] using Mathlib [24].

The catalogue reported here records the construction status of 18 September 2026: eight family packages and fifteen sporadic constructions, summarized in Formalization results 3.1 and 3.2. Representative constructions illustrate the mathematical approach. The full automorphism group of the split octonions gives $G_2(q)$ uniformly in all characteristics; singular-point geometry determines its order and proves simplicity for $q > 2$. Twisted seven-dimensional matrices give the small Ree groups, whose point geometry determines their full stabilizers and whose preserved tensors give embeddings into $G_2(q)$. The Conway–Parker algebra supplies the three Fischer simple groups through reflecting rays and commuting residues. In the classical families, explicit comparison isomorphisms relate models, while involution-class counts separate equal-order orthogonal and symplectic groups in odd characteristic and rank at least three.

Classification has two complementary components: constructing the groups on the list and proving that every finite simple group occurs there. ATLAS develops the first component

without assuming the exhaustiveness theorem of the classification of finite simple groups (CFSG). Construction, recognition, and exhaustiveness remain distinct obligations. The actions, stabilizers, and comparison maps retained here are intended as inputs to subsequent structure and recognition theorems; Section 2 states the conditional recognition questions precisely.

The mathematical sources combine established constructions with the author’s companion work. The Hall–Janko preprint [14] and the Conway–Parker algebra manuscript [13] supply source mathematics for the corresponding entries. The present article describes the models, their order and simplicity arguments, and the structural relationships preserved in their formal realization. The direct Monster-order work with Seysen [18] supplies a starting point for future formalization, not a completed entry in this release. Companion work on integral Albert algebras and the rank-26 lattice and Cayley-plane design develops the arithmetic and geometric direction [15, 17].

A broader objective is to explain why sporadic symmetries arise. Vertex operator algebras (VOAs) provide a principal direction, illustrated by the public Monster work in Section 10. Retaining the mathematical objects is essential for this purpose: the Fischer construction, for example, keeps its algebra, rays, and incidence geometry together with the arguments determining the group and its order. Formal verification makes these arguments and their structural maps available for checked reuse.

The name recalls the *Atlas of Finite Groups* [5]. Our initial scope is constructions and structural theorems; complete character tables and maximal-subgroup lists are longer-term projects. Section 4 records the classical conventions and comparison results, and Sections 5–9 describe representative constructions. Sections 10 and 11 distinguish further mathematical goals from the completed catalogue. Related formalization projects are discussed in Section 12. Technical implementation, declaration interfaces, logical foundations, and verification records are collected in Appendices A–D, including the complete per-group source-line dependency table.

2 The mathematical content of an entry

For a family label X , let $\mathcal{A}_X$ be an explicitly specified parameter domain. An entry consists of a defined group $G_X(\lambda)$, for $\lambda \in \mathcal{A}_X$, and proofs of

$$|G_X(\lambda)| = N_X(\lambda), \quad G_X(\lambda) \text{ is simple}, \quad P_X(G_X(\lambda)), \quad (2.1)$$

together with finiteness and, for the nonabelian entries, explicit noncommutativity. Here P_X describes independently defined mathematical structure: an action, a code, a lattice, an algebra, or specified maps between such objects. Neither $\mathcal{A}_X$ nor P_X may incorporate the desired order or simplicity assertion as an unproved premise. For cyclic groups of prime order, commutativity is part of the intended conclusion.

For some entries, we also compute $k_2(G)$, the number of conjugacy classes of nonidentity involutions, to distinguish nonisomorphic groups of the same order; see Section 4.4.

Here *construction* refers to a specified mathematical model and proofs of its properties. The development uses classical mathematics in Lean, including classical choice; it need not provide an executable enumeration of every group element.

2.1 The chosen object and its full symmetry group

For a symmetry-group model, equality between a generated subgroup and the full automorphism group is a mathematical theorem. In a code or lattice construction, generation or a stabilizer

argument establishes this equality before the order and simplicity assertions are attached to the full group.

Central quotients come with their maps and kernels. Thus Co_0 is distinguished from Co_1 , and a Hermitian isometry group from its scalar quotient. A different distinction is that between Fi_{24} and its derived subgroup Fi'_{24} . Abstract group isomorphisms, subgroup equalities, and equivariant identifications of actions are recorded with the corresponding data.

2.2 Conditional recognition and comparison

Conditional on CFSG, order determines a finite simple group up to isomorphism except for the pairs $\{\text{Alt}_8 \cong \text{PSL}_4(2), \text{PSL}_3(4)\}$ and $\{B_r(q), C_r(q)\}$ with $r \geq 3$ and q odd [20]. These pairs are distinguished by the number $k_2(G)$ of conjugacy classes of nonidentity involutions: the first has values 2 and 1, and the second r and $\lfloor r/2 \rfloor + 1$ [34, 4]. Consequently, assuming CFSG, the pair $(|G|, k_2(G))$ determines the isomorphism type of a finite simple group. The formal B/C counts and both nonisomorphism results are described in Section 4.4; the general recognition theorem and the $\text{Alt}_8/\text{PSL}_3(4)$ involution counts are proposed work.

For construction proofs, the relevant independence question concerns the hypotheses and dependency graph of each theorem. Background results and earlier constructions can be reused. A classification argument can also be non-circular when the property being deduced was not used in that argument, for example when it concerns smaller groups. The Monster order illustrates why precise attribution matters: Griess–Meierfrankenfeld–Segev, Corollary 3.7.3, uses substantial information about other sporadic groups [11], whereas Höhn–Seysen [18] gives a direct computational determination of the Monster and Baby Monster orders in the Leech-lattice setting. The former dependency is more specific than an invocation of the entire CFSG.

Recognition through an action or local geometry offers a natural interface: a classification argument produces the structure, and a comparison theorem identifies it with a constructed model. The actions, stabilizers, and maps retained here are intended to support such theorems.

3 Current constructions

The catalogue records eight completed family packages and fifteen sporadic constructions [12]. In this article, *completed* means that the stated order, simplicity, and structural results have the recorded proofs for the specified parameter domains; it does not mean that every property of the groups has been formalized. The verification scope is described in Appendix C. Family labels count construction packages and can overlap up to isomorphism. Section 4 records the status of the comparison maps.

Formalization result 3.1 (Family packages). *The following family packages are completed. Here n is the degree of an alternating group or the matrix dimension of PSL_n ; r denotes Dynkin rank in the B, C, D families. For each finite field F below, put $q = |F|$.*

- (i) *For every prime p , the cyclic group C_p has order p and is simple and commutative.*
- (ii) *For every $n \geq 5$, the alternating group Alt_n is nonabelian simple and has order $n!/2$.*
- (iii) *Let F be any finite field, put $q = |F|$, and let $n \geq 2$. The central quotient $\text{PSL}_n(F) = \text{SL}_n(F)/Z(\text{SL}_n(F))$ has order*

$$|\text{PSL}_n(F)| = \frac{q^{n(n-1)/2} \prod_{i=2}^n (q^i - 1)}{\gcd(n, q - 1)}. \quad (3.1)$$

It is simple if and only if $(n, q) \notin \{(2, 2), (2, 3)\}$. For those admissible parameters it is also nonabelian.

- (iv) For every finite field F and rank $r \geq 1$, both the projective quadratic model $B_r(F)$ and the symplectic quotient $C_r(F) = \mathrm{PSp}_{2r}(F)$ have the order in (4.1). Each is simple exactly outside $(r, q) = (1, 2), (1, 3), (2, 2)$, and nonabelian at the simple parameters. The full nonsimple models at $(r, q) = (2, 2)$ and their derived subgroups are distinguished. The symplectic rank-zero model is trivial.
- (v) For every finite field F and $r \geq 4$, the projective split-orthogonal model $D_r(F)$ has the order in (4.2) with $\varepsilon = 1$ and is nonabelian simple in every characteristic, including $D_4(2)$.
- (vi) For every finite field F , with $q = |F|$, let $G_2(F) = \mathrm{Aut}_F(\mathbb{O}_{\mathrm{split}}(F))$ be the full group of multiplication-preserving linear automorphisms of the constructed split octonion algebra. Then

$$|G_2(F)| = q^6(q^6 - 1)(q^2 - 1). \quad (3.2)$$

It is simple if and only if $q > 2$, and noncommutativity is proved for every q . At $q = 2$, an explicitly defined normal subgroup has order 6048 and index two; its own simplicity and identification with $\mathrm{PSU}_3(3)$ are not included in this result.

- (vii) Let F be a finite field of characteristic three with $|F| = q = 3^{2m+1}$, where $m \geq 1$. The group ${}^2G_2(F, m)$ generated by the Tits-twisted matrices in $\mathrm{GL}_7(F)$ is nonabelian simple and has order

$$|{}^2G_2(F, m)| = q^3(q^3 + 1)(q - 1). \quad (3.3)$$

It acts faithfully and doubly transitively on a constructed set of $q^3 + 1$ points. The full point stabilizer is the soluble Borel subgroup of order $q^3(q - 1)$. Field transport and an injective homomorphism into the split-octonion group $G_2(F)$ are proved. The parameter $q = 3$ and a graph-automorphism fixed-point characterization are not part of this assertion.

The orthogonal packages retain intrinsic kernels, full centers, root generation, singular-line geometry, and field transport. Their domains include comparison models outside the primary label ranges in Table 2. For type A and G_2 , finite-field witnesses give existence at every admissible prime power; for the small Ree groups, a canonical field witness is supplied for every $m \geq 1$. The notation ${}^2G_2(F, m)$ follows the Lean model's explicit twist parameter; $|F| = 3^{2m+1}$ determines m uniquely.

Cyclic and alternating entries reuse Mathlib's principal order and simplicity results. The remaining packages combine general library theorems with the constructions described below. Mathieu parameters such as a dodecad or marked coordinates select objects in constructed finite structures; standard choices are supplied.

Formalization result 3.2 (Sporadic packages). *The development contains finite nonabelian simple groups with the constructions and orders in Table 1. These entries include the specified code, lattice, stabilizer, scalar-quotient, or algebraic interfaces.*

The auxiliary groups, notably Co_0 and the retained scalar covers, are not counted as additional simple groups. Nor are small isomorphisms between family members counted as new entries. The eleven sporadics outside that completed catalogue are $J_1, J_3, J_4, \mathrm{He}, \mathrm{Ru}, \mathrm{O}'\mathrm{N}, \mathrm{HN}, \mathrm{Ly}, \mathrm{Th}, B$, and the Monster. The classical extension and the remaining uniform Lie-type programme are described next. The Tits group is a separate exceptional Lie-type entry, not an additional sporadic group.

Group	Order	Chosen construction
M_{11}	7 920	Point stabilizer in the dodecad model of M_{12}
M_{12}	95 040	Golay dodecad stabilizer
M_{22}	443 520	Pointwise stabilizer of two Golay coordinates
M_{23}	10 200 960	Stabilizer of one Golay coordinate
M_{24}	244 823 040	Full binary Golay code automorphism group
Co_1	4 157 776 806 543 360 000	Leech isometry group modulo $\{\pm I\}$
Co_2	42 305 421 312 000	Stabilizer of a Leech vector of squared norm 4
Co_3	495 766 656 000	Stabilizer of a Leech vector of squared norm 6
McL	898 128 000	Pointwise stabilizer of a Leech 2–2–3 triangle
HS	44 352 000	Pointwise stabilizer of a Leech 2–3–3 triangle
Suz	448 345 497 600	Full Eisenstein Hermitian group modulo six scalars
J_2	604 800	Full icosian Hermitian group modulo two signs
Fi'_{24}	1 255 205 709 190 661 721 292 800	Parity kernel in the constructed Fischer ray group
Fi_{23}	4 089 470 473 293 004 800	Centralizer of a marked involution modulo that involution
Fi_{22}	64 561 751 654 400	Common centralizer of two marked commuting involutions modulo their subgroup

Table 1: The fifteen completed sporadic constructions. A triangle label records half the three squared edge lengths. The groups are stabilizers of vectors or configurations, with the pointwise and setwise distinctions retained.

4 Classical groups and comparison isomorphisms

4.1 Current classical development and notation

The untwisted families A, B, C, D have completed packages. The unitary 2A and minus-orthogonal 2D families and a common Chevalley framework are further construction tasks.

For rank r over a finite field F , the orthogonal constructions start from

$$Q_B(x, y, z) = \sum_{i=1}^r x_i y_i + z^2, \quad Q_D(x, y) = \sum_{i=1}^r x_i y_i.$$

They construct the subgroups generated by Siegel transformations and divide by their scalar subgroups. In the completed parameter ranges, the elementary subgroups are identified in odd characteristic with the determinant/spinor kernels. In characteristic two, the B subgroup is the full quadratic isometry group and the split D subgroup is the intrinsic Dickson kernel. Their scalar subgroups are proved to be their full centers. The B geometric package has a faithful singular-line action in every positive rank, transitivity over finite fields, and primitivity from rank two; the split D package retains its faithful primitive singular-line action in rank at least four.

Throughout this section $q = p^f$ is a prime power, and r denotes *Dynkin rank*, not matrix dimension. We write Alt_m for an alternating group to distinguish it from the Lie-type symbol $A_r(q)$. The superscript in 2A_r denotes twisting; it is not a power of A_r . In twisted notation, q is the size of the fixed field: ${}^2A_r(q) = PSU_{r+1}(q)$ is constructed using a Hermitian space over $\mathbb{F}_{q^2}$. References writing ${}^2A_r(q^2)$ denote the same family with a different field convention. We use the fixed-field convention also for ${}^2D_r(q)$, ${}^2E_6(q)$, and ${}^3D_4(q)$.

The symbols denote central quotients. For even q , type B_r is realized on a nonsingular odd-dimensional quadratic space with one-dimensional polar radical; its symplectic comparison is proved. Exceptional small parameters require conventions for Ω as in Taylor [34, Chapters 8, 10–12].

Type	Classical model	Rank range	Exclusions for simplicity
$A_r(q)$	$\mathrm{PSL}_{r+1}(q)$	$r \geq 1$	$(r, q) = (1, 2), (1, 3)$
$B_r(q)$	$\mathrm{P}\Omega_{2r+1}(q)$	$r \geq 2$	$(r, q) = (2, 2)$
$C_r(q)$	$\mathrm{PSp}_{2r}(q)$	$r \geq 3$	None in this rank range
$D_r(q)$	$\mathrm{P}\Omega_{2r}^+(q)$	$r \geq 4$	None in this rank range
${}^2A_r(q)$	$\mathrm{PSU}_{r+1}(q)$	$r \geq 2$	$(r, q) = (2, 2)$
${}^2D_r(q)$	$\mathrm{P}\Omega_{2r}^-(q)$	$r \geq 4$	None in this rank range

Table 2: Classical families and proposed primary rank conventions, not the full implemented parameter domains. Types A, B, C, D have completed packages. The unitary and minus-orthogonal families remain to be constructed. The even-characteristic B/C comparison is proved for the retained models.

The proposed primary ranges $r \geq 1, 2, 3, 4$ for A, B, C, D remove low-rank repetitions while keeping the comparison models in the library. In particular, B, C include all positive ranks, and D_2, D_3 are available. The odd-characteristic B_2 construction uses its symplectic C_2 image, and the geometric proof uses transport through a B_1 complement. Thus the maps $B_1 \cong A_1$ and $B_2 \cong C_2$ connect useful constructions. Even-characteristic B/C isomorphisms and isolated coincidences still give overlaps between primary labels.

For reference, the classical order formulae are

$$\begin{aligned}
|A_r(q)| &= \frac{q^{r(r+1)/2}}{(r+1, q-1)} \prod_{i=2}^{r+1} (q^i - 1), \\
|{}^2A_r(q)| &= \frac{q^{r(r+1)/2}}{(r+1, q+1)} \prod_{i=2}^{r+1} (q^i - (-1)^i), \\
|B_r(q)| = |C_r(q)| &= \frac{q^{r^2}}{(2, q-1)} \prod_{i=1}^r (q^{2i} - 1), \tag{4.1}
\end{aligned}$$

$$|D_r^\varepsilon(q)| = \frac{q^{r(r-1)}(q^r - \varepsilon)}{(4, q^r - \varepsilon)} \prod_{i=1}^{r-1} (q^{2i} - 1), \tag{4.2}$$

where $(a, b) = \gcd(a, b)$, $D_r^+ = D_r$, and $D_r^- = {}^2D_r$ [34, 37]. The A, B, C and split- D formulae are proved for the models in the ranges stated above; the unitary and minus- D formulae are targets for subsequent development.

4.2 Uniform coincidences and low ranks

Table 3 records the uniform identifications of central quotients, including auxiliary models below the primary rank ranges. Here 2A_1 abbreviates PSU_2 ; type A_1 has no nontrivial Dynkin-diagram twist.

The exterior-square representation produces the six-dimensional orthogonal model from a four-dimensional linear space. Related Klein-correspondence arguments give low-rank comparisons. In characteristic two, quotienting the polar radical of an odd-dimensional quadratic space gives a symplectic space. Over a finite field this has a converse lift [34, Chapters 11–12]. The completed untwisted comparisons implement these constructions.

Isomorphism	Parameters and formalization status
$A_1(q) \cong B_1(q) \cong C_1(q) \cong \text{PSU}_2(q)$	$B_1 \cong A_1$ and $C_1 \cong A_1$ proved over every field; the PSU_2 link remains planned. Simple for $q \geq 4$.
$B_2(q) \cong C_2(q)$	Proved over every finite field, including the nonsimple $q = 2$ case.
$D_2(q) \cong A_1(q) \times A_1(q)$	Proved over every field; the split rank-two model is not a simple entry.
$D_3(q) \cong A_3(q)$	Proved over every finite field for the split six-dimensional model.
${}^2D_2(q) \cong A_1(q^2)$	All q mathematically; four-dimensional minus-type comparison still planned.
${}^2D_3(q) \cong {}^2A_3(q)$	All q mathematically; six-dimensional minus-type comparison still planned.
$B_r(2^f) \cong C_r(2^f)$	Proved for $r \geq 1, f \geq 1$; primary-label overlaps persist for $r \geq 3$.

Table 3: Uniform identifications and their current proof status. For $D_2(2)$ we retain the central quotient of the two $\text{SL}_2(2)$ factors, not a possibly different derived-full-orthogonal convention. The table includes comparison models outside the primary rank ranges.

4.3 Isolated coincidences across the classification list

Table 4 lists the isolated alternating and Lie-type coincidences, including simple derived groups at small parameters [5, 37, 38]. Its last column distinguishes proved maps from known isomorphisms awaiting formal comparison.

Order	Isomorphic simple groups	Status of comparison maps
60	$\text{Alt}_5 \cong A_1(4) \cong A_1(5)$	Both links with Alt_5 proved.
168	$A_1(7) \cong A_2(2)$	Proved through a common eight-point action.
360	$\text{Alt}_6 \cong A_1(9) \cong B_2(2)' \cong C_2(2)'$	All displayed links proved; the derived-group maps are explicitly exported.
504	$A_1(8) \cong {}^2G_2(3)'$	The $q = 3$ derived-subgroup package and this comparison remain to be formalized.
6 048	${}^2A_2(3) \cong G_2(2)'$	An index-two normal subgroup of $G_2(2)$ of order 6048 is constructed; its simplicity and the unitary comparison remain open in Lean.
20 160	$\text{Alt}_8 \cong A_3(2)$	Proved through the binary deleted permutation quadratic module.
25 920	${}^2A_3(2) \cong B_2(3) \cong C_2(3)$	$B_2 \cong C_2$ proved; the unitary link remains planned.

Table 4: Isolated cross-series coincidences and their formal status, up to the uniform identifications in Table 3. A prime denotes the derived subgroup. The status column records the available comparison maps.

The binary quadratic construction gives $B_2(2) \cong \text{Sym}_6$ and $B_2(2)' \cong \text{Alt}_6$. The full group has order 720 and its derived group order 360. The proved B_2/C_2 map identifies the full binary symplectic model, and its restriction to the derived subgroups is exported. The separate $\text{PSL}_2(9) \cong \text{Alt}_6$ comparison uses the projective line and the ten partitions of six letters into

two triples. Composing these proved maps completes the order-360 row.

For the other small parameters, $G_2(2)'$ has index two in $G_2(2)$, and ${}^2G_2(3)'$ has index three in ${}^2G_2(3)$. The Tits group ${}^2F_4(2)'$ has index two in ${}^2F_4(2)$ but gives no further coincidence in the table. The 26 sporadic groups supply no additional coincidences with each other or with the alternating and Lie-type simple groups [5, 37, 38]. Prime cyclic groups are separate; $\text{Alt}_3 \cong C_3$ lies outside the chosen alternating range. These tables do not classify nonsimple groups or central covers, and their exhaustiveness is not a Lean result here.

4.4 Equal orders and proved nonisomorphisms

The order collisions in Section 2 require *nonisomorphism* certificates, not additional comparison isomorphisms. Write $k_2(G)$ for the number of conjugacy classes of nonidentity involutions, not the number of involutions themselves. For every finite field of odd order and every positive rank, the formalized involution-class theorems give

$$k_2(B_r(q)) = r, \quad k_2(C_r(q)) = \lfloor r/2 \rfloor + 1. \quad (4.3)$$

Consequently, for $r \geq 3$ the constructed models satisfy

$$|B_r(q)| = |C_r(q)|, \quad B_r(q) \not\cong C_r(q).$$

The nonisomorphism follows from the isomorphism invariance of k_2 .

Equal-order pair	k_2 values	Formal status
$B_r(q), C_r(q)$	$r, \lfloor r/2 \rfloor + 1$	Order equality and nonisomorphism proved for odd $q, r \geq 3$.
$\text{Alt}_8, \text{PSL}_3(4)$	2, 1	Nonisomorphism proved via Sylow 2-subgroup centers; these k_2 counts are not certified.

Table 5: The exceptional equal-order pairs. In the second row, the k_2 values are known mathematically, while the formal nonisomorphism uses a different invariant. Conditional on CFSG these two kinds exhaust the nonisomorphic simple-group order collisions [20, 34, 4].

The proved order-20 160 obstruction constructs the upper unitriangular subgroups in $\text{SL}_4(2)$ and $\text{SL}_3(4)$ and embeds them in the corresponding projective groups. Both images have order 64 and are Sylow 2-subgroups; symbolic commutation formulas show that their centers have orders 2 and 4, respectively. An isomorphism of the ambient groups would preserve this invariant. Transport through the independently proved $\text{PSL}_4(2) \cong \text{Alt}_8$ map gives the alternating-group obstruction. The optional involution-class counts remain separate: Alt_8 has the two even cycle types $2^2 1^4$ and 2^4 , whereas $\text{PSL}_3(4)$ has one involution class [34, Chapter 4]. Both nonisomorphism routes are unconditional, but neither a complete catalogue-wide collision theorem nor the conditional recognition theorem for $(|G|, k_2(G))$ is claimed as a Lean result.

4.5 Comparison maps and remaining targets

The comparisons marked as proved in Tables 3 and 4 are explicit multiplication-preserving maps with proved inverses, rather than identifications by order. Field isomorphisms also induce isomorphisms of the corresponding orthogonal models.

The five isolated comparisons between the implemented families use natural actions or the deleted permutation quadratic module. Their order-60 argument also gives an elementary recognition theorem for nonabelian simple groups of that order. Unitary, twisted- D , and the remaining exceptional links are further targets. Each comparison of actions or central covers needs its own maps, kernels, and equivariance statement; the group equivalences alone concern the simple quotients.

5 A uniform example: projective special linear groups

The projective linear construction starts from matrix groups and central quotients, followed by parameterized generation and simplicity arguments. Its Mathlib inputs are recorded in Appendix A.

5.1 Order and the determinant quotient

The starting point is the standard count of ordered bases,

$$|\mathrm{GL}_n(q)| = \prod_{i=0}^{n-1} (q^n - q^i).$$

The scalar center of $\mathrm{GL}_n(q)$ has order $q - 1$. The center of $\mathrm{SL}_n(q)$ consists of the scalar matrices λI with $\lambda^n = 1$ and therefore has order $d = \gcd(n, q - 1)$, by cyclicity of $\mathbb{F}_q^\times$. These counts give (3.1). The development also proves a divisibility-safe identity

$$|\mathrm{PSL}_n(q)|(q - 1)d = \prod_{i=0}^{n-1} (q^n - q^i),$$

so that the natural-number division in the displayed order formula is known to be exact.

The determinant gives a stronger structural result. Over any field F and in positive dimension there is an isomorphism

$$\mathrm{PGL}_n(F)/\mathrm{PSL}_n(F) \cong F^\times / (F^\times)^n, \quad (5.1)$$

where $\mathrm{PSL}_n(F)$ on the left denotes its canonical image in $\mathrm{PGL}_n(F)$. Scaling a matrix changes its determinant by an n th power, so the map is well defined. If $\det(g) = u^n$, then $u^{-1}g$ has determinant one and represents the same projective transformation; this identifies the kernel. Surjectivity follows from that of the determinant. The formalized equivalence includes compatibility with matrix representatives. It is consequently usable independently of the finite order calculation.

5.2 Generation and simplicity

Elementary transvections are written $x_{ij}(a) = I + aE_{ij}$. The diagonal/transvection reduction available in Mathlib is completed by explicit two-coordinate identities, giving generation of $\mathrm{SL}_n(F)$ by transvections. For distinct i, j, k , the commutator identity

$$[x_{ik}(a), x_{kj}(b)] = x_{ij}(ab), \quad [g, h] = ghg^{-1}h^{-1}, \quad (5.2)$$

proves perfectness when $n \geq 3$. Generation and perfectness pass to the projective quotient.

For a line $L = Fv$, consider the subgroup of transformations

$$u_\varphi(x) = x + \varphi(x)v, \quad \varphi(v) = 0.$$

It is abelian, is normal in the stabilizer of L , and transforms covariantly under conjugation. Its projective images, as L varies, generate $\mathrm{PSL}_n(F)$. Together with the faithful primitive projective action, these facts fit the following criterion.

Proposition 5.1 (Iwasawa criterion). *Let a nontrivial perfect group G act faithfully and primitively on a set Ω with at least two points. Suppose that for some $\alpha \in \Omega$, the stabilizer G_α has an abelian normal subgroup A whose conjugates generate G . Then G is nonabelian simple.*

Proof. Let $1 \neq N \triangleleft G$. The N -orbits form a G -invariant partition. Faithfulness excludes the partition into singleton orbits for such N , and primitivity therefore makes N transitive. Hence $G = NG_\alpha$. In G/N , the image of every conjugate of A is consequently the image of A itself, because $A \triangleleft G_\alpha$. The conjugates generate G , so G/N is abelian. Perfectness implies $G/N = 1$, and $N = G$. A nontrivial perfect group cannot be abelian. $\square$

The existing formal criterion and projective-action infrastructure are reused. The added general-rank generation and perfectness arguments discharge its hypotheses. This proves simplicity for every field in dimension at least three. For finite fields in dimension two, the existing Mathlib theorem supplies simplicity when $q \geq 4$. The two exceptional cases are proved nonsimple using their projective actions and orders. The final family statement gives the exact exception set. Numerical examples follow by specialization.

6 The exceptional family $G_2(q)$ from split octonions

The first completed exceptional Lie-type family is constructed directly from the split octonion algebra. The formal model retains the algebra, its full linear automorphism group, and its singular-point geometry; see [37, Chapter 4] for the classical background. The uniform proof includes characteristics two and three.

6.1 The algebra and its full automorphism group

Write $\mathbb{O}_{\text{split}}(F) = \bigoplus_{i=1}^8 Fx_i$. The multiplication is the bilinear extension of the fixed integral split-octonion table used in the construction. In these coordinates, for $a = \sum a_i x_i$,

$$1 = x_4 + x_5, \quad \text{tr}(a) = a_4 + a_5, \quad N(a) = a_1 a_8 + a_2 a_7 + a_3 a_6 + a_4 a_5.$$

Conjugation is $\bar{a} = \text{tr}(a)1 - a$. The formal development proves both identity laws, bilinearity, the quadratic identity $a^2 - \text{tr}(a)a + N(a)1 = 0$, the conjugate-product identities, norm multiplicativity, and reversal of products under conjugation. These are polynomial identities over arbitrary commutative rings. The same algebra therefore applies in characteristic two.

The group $G_2(F)$ is the full subgroup of $\text{GL}(\mathbb{O}_{\text{split}}(F))$ preserving multiplication. Preservation of the identity, trace, norm, and conjugation is proved from this definition. For example, surjectivity forces the identity to be fixed; for a nonscalar a , comparison of its quadratic identity before and after applying an automorphism forces the trace and norm coefficients to agree. Thus the invariants follow from preservation of multiplication. The natural eight-dimensional representation is faithful, and coefficientwise field isomorphisms transport the group and its action.

6.2 Explicit subgroups, full stabilizers, and order

Six root transformations are defined and verified directly from the multiplication table. Their additive laws and parameter injectivity are proved. Ordered products give a bijection $F^6 \rightarrow U$ with explicit multiplication and inverse formulas. The diagonal torus T is identified with $F^\times \times F^\times$, normalizes U , and intersects it trivially. Consequently,

$$|U| = q^6, \quad |T| = (q-1)^2, \quad |UT| = q^6(q-1)^2.$$

Together with explicit Weyl transformations, these maps are proved to generate the full octonion automorphism group.

The singular vectors are the nonzero a with $\text{tr}(a) = N(a) = 0$. Their number is $q^6 - 1$. Indeed, after eliminating one coordinate using the trace equation, the count reduces to pairs of three-vectors and a scalar satisfying $u \cdot v = t^2$: for each $u \neq 0$ and each t , the v -fibre has q^2 elements; for $u = 0$, one has $t = 0$ and q^3 choices of v . Including zero gives q^6 vectors. The projective singular-point set $\mathcal{P}$ therefore has size

$$|\mathcal{P}| = \frac{q^6 - 1}{q - 1}.$$

Coordinate reductions prove transitivity and determine the full stabilizers. The pointwise stabilizer of the opposite vectors x_1, x_8 is isomorphic to $\text{SL}_2(F)$; the stabilizer of x_1 has a parameter bijection with $F^5 \times \text{SL}_2(F)$. The full stabilizer P of the line Fx_1 is the explicit parabolic, with a bijection

$$F^\times \times F^5 \times \text{SL}_2(F) \longrightarrow P.$$

This bijection is a parameterization of the stabilizer. Since $|\text{SL}_2(F)| = q(q^2 - 1)$, orbit-stabilizer gives

$$|G_2(F)| = \frac{q^6 - 1}{q - 1} (q - 1) q^5 q(q^2 - 1),$$

which is (3.2). The order proof is independent of perfectness, simplicity, and the binary exception.

6.3 Singular-point geometry and simplicity

The action on $\mathcal{P}$ is faithful. Its point stabilizer has four orbits, with lengths

$$1, \quad q(q + 1), \quad q^3(q + 1), \quad q^5. \quad (6.1)$$

Coordinate reductions identify these strata as orbits. Their geometry also proves primitivity in every characteristic, including $q = 2$. The block-size argument leaves only the projective points of $W = \langle x_1, x_2, x_3 \rangle$ as a possible nontrivial proper block. But the linear span of products $W \cdot W$ is the distinguished line Fx_1 . Any automorphism stabilizing W must therefore fix that point, contradicting transitivity of a block stabilizer on a nontrivial block.

At Fx_1 , a pair of commuting root groups gives an abelian subgroup $R_0 \cong (F^2, +)$ of order q^2 , normal in the full point stabilizer. Transport defines such a subgroup at every singular point, with independence of the chosen transporter and covariance under $G_2(F)$. For $q > 2$, explicit root and Weyl relations show that their conjugates generate the full group. The argument chooses a field element distinct from 0 and 1 and applies also when $q = 3$. Torus commutators then give perfectness. The faithful primitive action and these local abelian groups satisfy Proposition 5.1, proving simplicity. Noncommutativity is established separately by two explicit root maps, even at $q = 2$. The positive simplicity proof does not use the numerical order of $G_2(F)$.

6.4 The binary exception

For $F = \mathbb{F}_2$, let H be the normal closure of the long-root groups. The construction produces a second geometry: the two-dimensional singular subspaces on which all products vanish. There are exactly 63 such zero-product planes. The count comes from an exhaustive check of 64×64 binary coordinate pairs; the group acting is still the full octonion automorphism group.

Two root involutions fix respectively 7 and 9 of these planes, so their permutation signs are $(-1)^{28} = 1$ and $(-1)^{27} = -1$. The resulting sign homomorphism is surjective and contains

H in its kernel. Independently, primitivity of the singular-point action makes the nontrivial normal subgroup H transitive, and the explicit point-stabilizer generators give $[G_2(2) : H] \leq 2$. It follows that H is exactly the sign kernel. Using the already proved order,

$$|G_2(2)| = 12096, \quad [G_2(2) : H] = 2, \quad |H| = 6048.$$

Field transport gives the same conclusion for every field of cardinality two. This proves the negative case in the exact simplicity theorem. The present package does not prove $H = G_2(2)'$, simplicity of H , or its identification with $\text{PSU}_3(3)$; the conventional derived-group coincidence in Table 4 remains a separate comparison target.

7 Small Ree groups from twisted matrices

Let F be a finite field of characteristic three with $q = |F| = 3^{2m+1}$ and $m \geq 1$. The model is the subgroup of $\text{GL}_7(F)$ generated by three explicit root maps, a diagonal torus, and a Weyl involution. The coordinates follow B  arnhielm's explicit matrix model [1, Section 3]; the geometric and simplicity arguments draw on Wilson's elementary constructions [36, 35]. Its order and simplicity are proved for this generated group, without an abstract Ree-group theorem or recognition by order.

7.1 Field twists and the root group

The twisting maps are the iterated Frobenius automorphisms

$$\theta(x) = x^{3^m}, \quad \sigma(x) = x^{3^{m+1}}.$$

The finite-field identity $x^{3^{2m+1}} = x$ gives $\theta\sigma = \sigma\theta = \text{id}$ and $\sigma^2(x) = x^3$. These identities, their compatibility with field equivalences, and the bound $q \geq 27$ are proved before the group arguments.

Write $S(a, b, c)$ for the product of the three root matrices in their specified order. Its entries recover a, b, c , and direct matrix multiplication gives

$$S(a, b, c)S(d, e, f) = S(a + d, b + e - a\theta(d)^3, c + f - db + a\theta(d)^3d - a^2\theta(d)^3).$$

Together with the inverse formula, this identifies the root subgroup U with the parameter set F^3 , so $|U| = q^3$. Projection to the first parameter has abelian image and abelian kernel, and hence U is soluble. The diagonal subgroup $H \cong F^\times$ normalizes U , with conjugation on parameters

$$(a, b, c) \mapsto \left(\frac{a\theta(\lambda)^3}{\lambda^2}, \frac{b\lambda}{\theta(\lambda)^3}, \frac{c}{\lambda} \right).$$

Unique factorization in $B = UH$, the equality $U \cap H = 1$, and the split projection $B \rightarrow H$ with kernel U prove that B is soluble and has order $q^3(q - 1)$.

7.2 The point geometry and the full stabilizer

The Ree–Tits point set consists of a point at infinity and the projective classes of the first rows of the matrices $S(a, b, c)$. Projective parameter injectivity gives $q^3 + 1$ points, and U acts regularly on the affine points. The matrix convention uses row vectors; inverse right multiplication supplies the corresponding left action.

The main geometric step is to prove preservation by the Weyl element and to identify the full stabilizer of infinity. Seven exterior-kernel relations and seven Tits-twisted bilinear equations characterize the point chart. Their invariance under negative coordinate reversal proves Weyl preservation. They also imply that the last affine coordinate vanishes only at the affine origin. The generated group is consequently doubly transitive.

Every generator preserves an explicit alternating cross product and the twisted bilinear relation. At infinity the relation recovers a distinguished two-dimensional subspace, and the cross product recovers the full flag. A matrix fixing infinity must therefore be triangular. Applying the opposite flag at the affine origin shows that the two-point stabilizer is diagonal; the invariant tensors force it to be exactly H . Regularity of U then gives the full one-point stabilizer B . This proves faithfulness as well as the order formula (3.3) by orbit-stabilizer. The stabilizer calculation completes the order proof.

7.3 Perfectness, simplicity, and comparison with octonions

For $q \geq 27$, torus commutators place all three root maps in the derived subgroup. Short characteristic-three matrix identities then place the Weyl involution and the minus-one torus element there. Weyl inversion supplies all torus squares. Since $q \equiv 3 \pmod{4}$, every nonzero scalar is a square or the negative of a square, so the whole torus lies in the derived subgroup. The group is therefore perfect.

Simplicity uses a separately proved general criterion: a nontrivial perfect group with a faithful quasiprimitive action and a soluble point stabilizer is simple. Indeed, any nontrivial normal subgroup N is transitive, so the stabilizer surjects onto G/N . This quotient is both soluble and perfect and is therefore trivial. The Ree action is primitive, and its full stabilizer is the soluble subgroup B . Explicit noncommuting root elements supply noncommutativity. This simplicity argument does not use the numerical group-order theorem.

Entrywise field equivalence induces a group isomorphism of the models. A separate comparison embeds the group into the split-octonion $G_2(F)$ of Section 6. In the fixed seven-coordinate basis, the scalar-vector identification with the eight-coordinate octonions is

$$(t, v) \mapsto (v_0, v_1, v_2, t - v_3, t + v_3, -v_4, -v_5, -v_6).$$

In characteristic three the octonion product becomes

$$(s, v)(t, w) = (st - B(v, w), sw + tv + v \times w),$$

where B is the preserved symmetric form and $\times$ the preserved cross product. Extending inverse row multiplication while fixing the scalar coordinate thus gives an injective homomorphism into the full octonion automorphism group. This uses structural octonion results, not the order or simplicity of $G_2(F)$. No equality with the fixed points of an exceptional graph automorphism is asserted.

Specializations of the uniform order theorem at $q = 27, 243, 2187$ give, respectively,

$$10\,073\,444\,472, \quad 49\,825\,657\,439\,340\,552, \quad 239\,189\,910\,264\,352\,349\,332\,632.$$

These values specialize the uniform theorem to constructed finite fields. The public family excludes $q = 3$; the derived-group comparison with $\mathrm{PSL}_2(8)$ in Table 4 remains a separate task.

8 Sporadic groups from retained structures

The sporadic branch follows connected constructions. The code used to construct the Mathieu groups is retained in the Leech lattice, and the resulting lattice and its full symmetry group

are retained in later stabilizer and scalar-centralizer constructions. Standard background for these relationships is provided by [6]; the formal models and recorded completion results are those of the ATLAS development [12].

8.1 Trio code, hexacode, and Golay code

The initial construction passes from a quadratic alphabet and its trio code to the additive Kleinian hexacode, and then to the binary Golay code by two explicit twists. The coordinate organization

$$3 \longrightarrow 3 \times 2 \longrightarrow (3 \times 2) \times 4$$

is retained along with markings, partitions, and recovery maps. The quadratic functions on the intermediate alphabets are part of the data. Any comparison with an $\mathbb{F}_4$ -linear hexacode is a separate mathematical statement.

The constructed binary code C is a doubly even self-dual $[24, 12, 8]$ code. Its weight-eight supports give the Witt design $S(5, 8, 24)$. The chosen model for M_{24} is the full coordinate automorphism group of this code, and equality with the automorphism group of its octad design is proved in the same ambient symmetric group. Sextet geometry supplies local structure: every tetrad has a unique sextet completion, there are 1771 sextets, and the full sextet stabilizer has order 138240. The subsequent global action arguments give order 244823040, five-transitivity, and simplicity.

The smaller Mathieu groups come from stabilizers. Fixing one or two coordinates gives M_{23} and M_{22} ; stabilizing a dodecad and then one of its points gives M_{12} and M_{11} . Their actions on complements or dodecads, the corresponding residual designs, and the embeddings relating these choices are part of the construction. In particular, a pointwise two-coordinate stabilizer is kept distinct from the larger setwise stabilizer.

8.2 The Leech lattice and Conway groups

The retained Golay code defines an integral lattice Λ . Its rank 24, evenness, positive definiteness, unimodularity, minimum squared norm 4, and generation by minimal vectors are proved from the construction. The group Co_0 is its full linear isometry group, and

$$\text{Co}_1 = \text{Co}_0 / \langle -I \rangle.$$

Finiteness and the relevant kernels are proved for these groups before the final simplicity arguments.

The shell counts for squared norms 4, 6, 8 are respectively

$$196560, \quad 16773120, \quad 398034000.$$

They arise from exhaustive coordinate descriptions of lattice vectors. The global order proof uses minimal vectors and orthogonal pairs. An explicitly constructed nonmonomial isometry, together with the full monomial subgroup, generates Co_0 . The resulting order is

$$|\text{Co}_0| = 8\,315\,553\,613\,086\,720\,000.$$

The action on the 8292375 intrinsic crosses, and its primitivity, follow from the minimal-vector and orthogonal-pair analysis.

Vector stabilizers produce Co_2 and Co_3 . Pointwise triangle stabilizers produce McL and HS . These models retain their embeddings in the lattice isometry group. Additional geometry includes the 276-point action of Co_3 and the rank-three 100-point graph action of HS . The lattice representations and stabilizer descriptions allow these groups to be reused without reconstructing unrelated permutation models.

8.3 Eisenstein and icosian structures

The constructions of Suz and J_2 enrich the same Leech lattice with additional scalar structure. They illustrate why integral comparison maps matter: a rational isometry between two ambient spaces must be shown to map the lattices onto one another. The documented comparisons prove both integral containments.

For Suz, a constructed fixed-point-free isometry ρ satisfies

$$\rho^3 = I, \quad \rho^2 + \rho + I = 0.$$

Its full centralizer $C_E = C_{\text{Co}_0}(\rho)$ is identified with the full Hermitian isometry group of the corresponding Eisenstein lattice. Its center consists of six scalar isometries. The constructed simple quotient has order

$$|C_E|/6 = 2\,690\,072\,985\,600/6 = 448\,345\,497\,600.$$

The action on 232960 intrinsic frames and the thirteen suborbits of a frame stabilizer are proved from the lattice. The sign quotient $C_E/\{\pm I\}$ embeds into Co_1 and maps onto Suz with central kernel of order three. This keeps the embedded triple extension distinct from its simple quotient.

For J_2 , let $\mathcal{I}$ be the constructed icosian order over $\mathbb{Z}[(1 + \sqrt{5})/2]$, and let C_H be the full quaternion-linear Hermitian isometry group of the associated rank-three congruence lattice. The underlying integral lattice is identified with the retained Leech lattice. The 120 norm-one icosian units act by right scalars, and C_H is their full centralizer in Co_0 . The scalar action and the center of C_H are separate objects.

There are 37800 vectors of quaternionic Hermitian norm two, and each associated right quaternionic line contains exactly 120 of them. Thus there are 315 root lines. Each is orthogonal to ten others, and every orthogonal pair has a unique completing root line. This gives an intrinsic count of frames,

$$\frac{315 \cdot 10}{6} = 525.$$

The full coordinate-frame stabilizer has order 2304. Explicit quaternionic reflections generate the full group and prove frame transitivity, whence

$$|C_H| = 525 \cdot 2304 = 1209600, \quad |C_H/\{\pm I\}| = 604800. \quad (8.1)$$

The point stabilizer has suborbits of sizes 1, 10, 32, 32, 80, 160. Their proved geometry yields primitivity. Reflection relations yield perfectness, and the cyclic subgroup generated by a point reflection supplies the abelian normal subgroup in Proposition 5.1. The sign quotient is therefore nonabelian simple. This example exhibits the chain from arithmetic and local geometry to a full symmetry group, its order, and its simplicity.

9 The Fischer constructions

The development contains concrete models and proofs of order, simplicity, finiteness, and noncommutativity for Fi'_{24} , Fi_{23} , and Fi_{22} . Their construction packages also retain the algebra, reflecting rays, and residue geometry described below.

9.1 The algebra, the ray group, and its residues

The starting point is the explicit Conway–Parker coordinate algebra constructed from Golay and Parker data [13]. The algebra has dimension 783 over $E = \mathbb{Q}(\omega)$, where ω is a primitive

cube root of unity; its coordinate index is the disjoint union of the 24 Golay coordinates and the 759 octads. A proved tensor identity supplies the reflecting automorphisms. Its normalized μ_3 -rays are three-element scalar orbits. Let F be the subgroup of permutations of the displayed reflecting rays generated by the constructed root involutions. The simple model Fi'_{24} is the kernel of its parity homomorphism. This kernel is proved equal to F' , with $[F : F'] = 2$.

For marked commuting basic involutions t_i , the residue construction uses the centralizer of the marked set and divides by the subgroup it generates. In particular, the two fixed models are

$$G_{23} = C_F(t_i)/\langle t_i \rangle, \quad G_{22} = C_F(t_i, t_j)/\langle t_i, t_j \rangle \quad (i \neq j).$$

The chosen marked involutions are part of the construction. The quotient kernels have orders two and four, respectively. Both residue models are therefore specified group quotients.

The order and simplicity assertions are proved for these defined models; neither assertion is supplied as a hypothesis.

The full ray group F has a primitive rank-three action with subdegrees 1, 31671, 275264. For its parity kernel Fi'_{24} , the proved action on the same 306936 rays is faithful and transitive; the recorded result does not include a rank-three assertion for this subgroup. The residue groups G_{23} and G_{22} have faithful primitive rank-three actions with point counts and subdegrees

	degree	subdegrees		
G_{23}	31671	1	3510	28160
G_{22}	3510	1	693	2816

The residue projections, their surjectivity, generation by distinguished involutions, and perfectness are also retained. These give structural interfaces for the residue groups.

The completion record also retains the nonsplit central double cover of G_{22} inside G_{23} and the perfect central triple cover of Fi'_{24} acting faithfully on the 783-dimensional E -algebra. The quotient's action is projective; no ordinary representation of the same degree is inferred for it. In the full semilinear group, odd symmetries invert the cubic scalars, so that scalar kernel is not central.

9.2 Ray counts and group orders

The finite-algebra construction produces three families of rays of sizes

$$24, \quad 759 \cdot 2^5, \quad \binom{24}{2} 2^{10},$$

whose sum is 306936. Frame geometry yields the largest group, and commuting residues yield the two smaller ones. The notation distinguishes the three-transposition group Fi_{24} from its derived subgroup Fi'_{24} , the sporadic simple group. A central triple cover, when retained at the algebra level, is a further distinct group. The factorizations of the orders proved for the formal models are

$$\begin{aligned} |\text{Fi}_{22}| &= 2^{17} 3^9 5^2 7 \cdot 11 \cdot 13, \\ |\text{Fi}_{23}| &= 2^{18} 3^{13} 5^2 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23, \\ |\text{Fi}'_{24}| &= 2^{21} 3^{16} 5^2 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29. \end{aligned}$$

The factor of two between $|\text{Fi}_{24}|$ and $|\text{Fi}'_{24}|$ is part of the derived-subgroup analysis.

10 The Monster and vertex operator algebras

Höhn–Seysen [18] determines the Monster and Baby Monster orders directly from computations with axes in the 196884-dimensional Griess algebra, using Seysen’s `mmgroup` implementation [31]. The Monster is generated by a subgroup of shape $2_+^{1+24}.\text{Co}_1$ and a triality element. Orbit and stabilizer calculations use the Golay code, Leech lattice, and Conway groups.

VOA arguments complement these calculations. The axes correspond, after normalization, to Ising vectors in the Moonshine module $V^\natural$. Orbifold and fusion arguments control involutions and their centralizers, while the framed VOA structure identifies $\text{Aut}(V^\natural)$ with the automorphisms of the Griess algebra preserving its invariant bilinear form. Combining the computed order with results of Carnahan and Borcherds, the paper proves that the constructed Monster is this full automorphism group and has exactly two conjugacy classes of involutions [18, Sections 2 and 5]. These are public mathematical results supporting future formalization; the Monster and Baby Monster are among the outstanding ATLAS entries.

This work continues the Moonshine programme of Borcherds [2] and Frenkel–Lepowsky–Meurman [7]. Borcherds’s proof of the Monstrous Moonshine conjectures [3] is a longer-term formalization goal, not a result of the present development. The Baby Monster vertex operator superalgebra provides a related source of sporadic symmetry [16]. Further examples include the 10-dimensional quadratic space over $\mathbb{F}_2$ recovered from the modules of $V_{\sqrt{2}E_8}^+$ in Shimakura’s treatment, with automorphism group $O_{10}^+(2)$ [32, Theorem 4.5], and Griess–Lam’s VOA existence proof of the Monster [10].

11 Further development

Completion of the full sporadic catalogue is a longer-term objective. The remaining eleven sporadic groups in Section 3 call for geometric constructions or verified computational arguments; structural and comparison theorems for completed entries are a separate continuing task. The Hall–Janko, Fischer, Albert-algebra, and Cayley-plane source manuscripts develop the geometric direction. For O’Nan’s group, Lee’s 2025 account records the absence at that time of a computer-free construction [23]; Jansen–Wilson give explicit modular constructions [19]. Formalizing such arguments requires proofs of the algorithms and checks of the finite data they use.

A common Lie-type framework should construct groups from root data, integral forms, and root operators, then establish generation, Bruhat decomposition, centers, orders, and simplicity. Steinberg [33] and Geck [8] provide sources. Beyond the completed families, the targets are the unitary and minus-orthogonal families, F_4, E_6, E_7, E_8 , and the graph-field twists 2E_6 and 3D_4 . The Suzuki and large Ree families

$${}^2B_2(2^{2e+1}), \quad {}^2F_4(2^{2e+1}) \quad (e \geq 1)$$

require exceptional characteristic-two endomorphisms. Their initial nonsimple parameters and the Tits group ${}^2F_4(2)'$ need separate statements [37]. Comparison with the existing G_2 and small Ree models, including the exceptional fixed-point description of the latter, is part of this programme.

For integration with classification arguments, priorities include centralizers of prime-order elements, Sylow and local subgroups, outer automorphisms, central extensions, and recognition geometries. The comparison maps of Section 4 allow these results to pass between the models encountered in different proofs.

Formalizing the Monster and Baby Monster results discussed in Section 10 requires both verified finite computations and infrastructure for graded spaces, vertex operations, lattice

extensions, and the relevant subalgebras or commutants. These developments would connect the finite-group constructions to their vertex-algebraic sources.

12 Related formalization projects

The Odd Order project of Gonthier and collaborators formalized the Feit–Thompson theorem in Coq (now Rocq), using Mathematical Components [9]. Completed in 2012 and described in 2013, it proves that finite groups of odd order are soluble and supplies extensive reusable finite-group and representation theory. Mathlib supplies the cyclic, alternating, and rank-two PSL results used here, together with group-action and simplicity infrastructure [24, 25].

The Qiuzhen-CFSG project and FormaTheoria workflow report a Lean development whose public README lists the Odd Order and Bender–Suzuki theorems as completed [29]. Their reconstruction of interdependent theories from the literature complements the construction work here. This account follows the authors’ public description and does not report an independent rebuild of their project.

Two Utrecht theses from 2024 precede this work: Roxy van de Kuilen’s partial formalization of the first Janko group [22] and Erik van der Plas’s study of the Conway groups in Lean [30]. Their precise declaration interfaces merit comparison with the constructions presented here.

The public repository `KitaKen1/finite-simple-groups-lean` reports kernel-checked order and simplicity certificates for fourteen sporadic permutation groups [21]:

$$M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, \text{HS}, J_2, J_1, \text{McL}, \text{Co}_3, \text{Suz}, \text{Co}_2, \text{Fi}_{22}, \text{He}.$$

Its README also lists nine individual Lie-type groups, including $G_2(4)$ and $\text{PSL}_3(4)$. External programs generate candidate certificates whose required properties are checked in Lean. This description follows the public repository consulted on 18 September 2026, without an independent audit of its proofs.

Parameterized families require constructions and proofs for all admissible ranks and fields. Root-group or geometric arguments provide one route; a uniformly verified algorithm could provide another. For individual groups, comparison isomorphisms could connect the permutation models with the code, lattice, and algebra models here, allowing their structural results to be combined.

A Lean implementation and mathematical interfaces

A.1 Lean and Mathlib inputs

Lean supplies dependent type theory, elaboration, and the proof kernel; Mathlib supplies mathematical definitions, theorems, and proof-producing automation. In particular, `Group`, `Nat.card`, and `IsSimpleGroup` are defined in `Mathlib.Algebra.Group.Defs`, `Mathlib.SetTheory.Cardinal.Finite`, and `Mathlib.GroupTheory.Subgroup.Simple`. Appendix B records their interpretation and the logical axioms used.

Table 6 lists representative mathematical inputs. Declaration dependencies identify the portion used by a particular proof; file imports make a broader interface available.

The cyclic entry uses `ZMod.card` and `isSimpleGroup_of_prime_card`; the alternating entry uses `nat_card_alternatingGroup` and `alternatingGroup.isSimpleGroup`. For PSL, the existing `rank_two_simple` theorem and `MulAction.IwasawaStructure.isSimpleGroup` support the general-rank arguments in Section 5. ATLAS develops the octonion multiplication

Representative modules (prefix <code>Mathlib.</code> omitted)	Mathematical input
<code>Data.Fintype.Card</code> <code>SetTheory.Cardinal.Finite</code> <code>GroupTheory.Index</code>	Finite types, cardinalities, subgroup indices, and counting identities.
<code>GroupTheory.QuotientGroup.Basic</code> <code>GroupTheory.Subgroup.Centralizer</code> <code>GroupTheory.Commutator.Basic</code>	Quotient groups, centralizers, commutators, and the maps used in central and derived-group constructions.
<code>GroupTheory.GroupAction.Primitive</code> <code>GroupTheory.GroupAction.MultipleTransitivity</code> <code>GroupTheory.GroupAction.Iwasawa</code>	Primitive and multiply transitive actions and the Iwasawa simplicity criterion.
<code>GroupTheory.SpecificGroups.Cyclic</code> <code>GroupTheory.SpecificGroups.Alternating</code> <code>GroupTheory.SpecificGroups.Alternating.Simple</code>	Existing cyclic and alternating group theory, including principal order and simplicity results.
<code>Data.ZMod.Basic</code> <code>FieldTheory.Finite.GaloisField</code>	Modular arithmetic and finite fields of prime-power order.
<code>LinearAlgebra.Matrix.GeneralLinearGroup.Card</code> <code>LinearAlgebra.Matrix.SpecialLinearGroup</code> <code>LinearAlgebra.Matrix.Transvection</code>	Matrix groups, general-linear-group cardinalities, and elementary matrices.
<code>LinearAlgebra.Matrix.ProjectiveSpecialLinearGroup</code> <code>LinearAlgebra.Projectivization.Action</code> <code>LinearAlgebra.Projectivization.PSL.PSL2</code>	Projective groups and actions, including the existing dimension-two simplicity theorem.
<code>LinearAlgebra.Basis.Basic</code> <code>LinearAlgebra.BilinearForm.DualLattice</code> <code>LinearAlgebra.QuadraticForm.Basic</code> <code>Algebra.Quaternion</code>	Bases, bilinear and quadratic forms, dual lattices, and quaternion arithmetic underlying the geometric constructions.

Table 6: Representative mathematical inputs from the pinned Mathlib dependency; they are not an exhaustive or minimal import list.

and stabilizers, Golay–Leech structures, and Conway–Parker algebra on this foundation. Automation such as `ring` and `norm_num` produces proof terms checked by the kernel.

Shared mathematics can occur below public group interfaces. For example, `McL` and `HS` use `Atlas.Sporadic.Conway3.card`, while the Fischer entries share algebra and residue lemmas even when another Fischer group’s principal wrapper is absent from their dependencies. The source-line footprints in Appendix D count ATLAS declarations; their derivations also use Lean and Mathlib. The pinned environment is recorded there.

A.2 Principal declaration interfaces

The following identifiers connect the mathematical models described in Sections 4–9 to their formal statements. They are collected here to keep the construction arguments independent of implementation-level notation.

Classical comparisons. The comparisons in Tables 3 and 4 are Lean `MulEquiv` objects: multiplication-preserving maps with proved inverses. The classical index exports `b1EquivA1`, `b2EquivC2`, `d2EquivProduct`, `d3EquivA3`, and `evenBEquivC` under `Atlas.Comparisons.Classical`. The symplectic rank-one map is `Atlas.Symplectic.rankOnePSL`; field equivalences induce `bFieldEquiv` and `dFieldEquiv`. The binary comparisons are `Atlas.Orthogonal.b2BinaryEquivSymmetric` and `Atlas.Orthogonal.b2BinaryDerivedEquivAlternating`. The theorem

`Atlas.Comparisons.Classical.oddB_same_order_nonisomorphic_C` records equal order and the nonexistence of a `MulEquiv`, using the conjugacy-class invariance of k_2 .

Split octonions. The model `Atlas.G2.Model` uses a generic construction of multiplication-preserving linear automorphisms that requires no associativity assumption. The principal declarations in `Atlas.G2` are `card_Model`, `isSimple`, and `isSimple_iff`, with `exists_mul_ne_mul` for noncommutativity. The public `Atlas.typeG2_construction` package collects the results for the concrete model, and `Atlas.typeG2_prime_power` supplies finite fields for every prime power. The corresponding existence endpoints retain the concrete model as their witness. The evidence for order, positive simplicity, the binary exception, and the complete interface is separately scoped in Appendix C. The exhaustive binary coordinate-pair check in Section 6 uses ordinary kernel reduction.

Small Ree groups. The model is `Atlas.ReeG2.ModelFm`. The principal declarations are `Atlas.ReeG2.order` and `Atlas.ReeG2.simple`. The public package `Atlas.typeReeG2_construction` and existence theorem `Atlas.exists_typeReeG2_parameter` use the same model.

Fischer groups. The declaration `Atlas.Fischer.rootGeneratedRayGroup` defines the ray group F of Section 9. The `derived` and `index` theorems identify the parity kernel with F' and prove $[F : F'] = 2$. For each Fischer entry, the public declarations `card`, `simple`, `finite`, and `exists_mul_ne_mul` establish the corresponding basic assertions. The expanded signatures contain no supplied order or simplicity hypothesis. The namespaces under `Atlas.Sporadic` are `Fischer24Prime`, `Fischer23`, and `Fischer22`. Construction and existence interfaces accompany these results. The retained extension statements have their own statement and dependency certificates.

B Formal statements and logical foundations

For a constructed carrier G , `Group G` supplies operations and proofs of the group laws. Subgroups and quotients obtain these from their ambient groups and proved normality. On finite types, `Nat.card G` is the number of elements; it is zero on infinite types. Finiteness is also exported as `Finite G`. `IsSimpleGroup G` asserts nontriviality and that every normal subgroup is trivial or the whole group. For the nonabelian entries, noncommutativity is proved separately.

For example, the fixed model `Atlas.Sporadic.Fischer22.Model` is the centralizer quotient G_{22} of Section 9. Its public statements are

```
finite :      Finite G22,
card :      Nat.card G22 = 64561751654400,
simple :      IsSimpleGroup G22,
exists_mul_ne_mul :  ∃x, y ∈ G22, xy ≠ yx.
```

All four declarations lie in `Atlas.Sporadic.Fischer22`; the display suppresses printer-level universe and typeclass arguments.

The recorded dependency audits report only `propext` (propositional extensionality), `Classical.choice`, and `Quot.sound` (equality of related quotient representatives). They exclude `sorryAx`, project-specific mathematical axioms, and native-evaluator trust shortcuts. The proofs have been accepted by Lean’s standard kernel and, for the complete exported dependency closure of the release selection, by the independently implemented Nanoda checker (Appendix C.2). These checks still rely on the foundations, the checking implementations, the

export pipeline, and the host system. The axiom report must be read together with the full theorem type and the definitions of its carrier, parameters, and instances: an axiom check alone does not establish that the statement expresses the intended mathematics [26, 27].

C Development and recorded verification

C.1 Mathematical specification and implementation

The author specifies the objects, theorems, proof architecture, and permitted dependencies. Codex, OpenAI’s coding agent, searches the pinned library and implements and repairs proofs against Lean’s elaborator and kernel. The author used ChatGPT, including GPT-6 Pro, for mathematical discussion and specifications, and Codex with GPT-6 Astra for Lean implementation. These names identify the development tools used; they are not a controlled comparison of models or a measurement of autonomous performance.

The author directs and maintains ATLAS and is responsible for its mathematical claims and formal specifications. Generative AI also assisted the drafting and revision of this article; the author is responsible for its final wording and mathematical content. Mathematical sources and joint work are credited where used. Compiled statements and dependency reports support review of the results; reproducing the AI’s search is unnecessary for checking the resulting proof terms.

C.2 Recorded verification

The completed runs of 18 September 2026 concern release commit 935474b and a selection of 911 declarations covering catalogue roles, structural properties, and comparison interfaces for the eight families and fifteen sporadic entries. The selection counts declarations, not groups. The checks address the formal statements, their dependencies, and acceptance by two checker implementations; the precise source state and checking details are given in Appendix D.

Build and dependency audit. The project-source build completed successfully, with 5 677 Lake jobs. The compiled-statement and transitive dependency audit covered 911 roots and 70 266 nodes, including type and proof dependencies, and found only the three permitted logical axioms of Appendix B. The build used pinned library caches, rather than recompiling every Mathlib source file.

Independent proof replay. Nanoda, an independently implemented checker written in Rust, accepted all 70 891 declarations in the complete exported dependency closure of the 911 selected roots, with no errors. This includes the library dependencies required by those roots, not unrelated imported theorems or all of Mathlib. Nanoda’s exported-declaration count and the dependency audit’s node count are produced by different tools and are reported separately.

Comparator. The same 911 targets passed Comparator: 869 theorem targets and 42 non-theorem targets represented by equality wrappers, with no definition holes. Statement and definition comparisons, permitted-axiom checks, standard Lean kernel replay, and the quotient post-check all succeeded. The Challenge and Solution import the same frozen release. This checks agreement with that formal reference and replay acceptance, rather than comparison against an independently authored mathematical specification. Comparator used the standard Lean kernel; Nanoda was a separate run.

Statement consistency. Additional inspections of the compiled order and simplicity statements confirmed that each pair uses the same model, parameters, and inherited group law. For the eight families, admissibility predicates contain the intended numerical restrictions rather than assumed conclusions, and q is the cardinality of the same field F wherever field parameters occur. The Lie-rank and matrix-dimension conventions agree. For the fifteen sporadic entries, the inspections also checked Mathieu markings and the distinctions between stabilizers, simple quotients, covering groups, and the Fi'_{24} parity kernel. These are inspections of the carrier applications and hypotheses, not additional kernel runs or external recognition theorems.

The earlier development catalogue check validated 886 declaration references, including aliases and structural exports. The source-footprint measurement used a separate selection of 46 principal order and simplicity declarations for 23 entries, with a complete graph of 65 676 dependency nodes and project source ranges recovered throughout. Table 7 retains that measurement. Neither the broader release audit nor the replay counts replace this line-count calculation. Family, geometric, and comparison audits use their own selections.

In particular, the G_2 and small Ree order audits exclude their perfectness and simplicity arguments. Their positive simplicity audits exclude the numerical group-order theorems. The Ree proofs also exclude untwisted G_2 results; its octonion embedding uses a separate structural scope. Finite tables checked by ordinary kernel reduction are part of some proofs.

C.3 Repository and checking instructions

The source repository is <https://github.com/Moonshine-in-Kansas/constructive-atlas>. The release catalogue is `docs/catalogue.json`; the build and dependency-audit instructions are `AUDIT.md` and `verification/check.rb`. Appendix D gives the tool revisions, checking commands, and scope and resource use of the completed runs.

The release’s default license is the *ATLAS Research and Attribution License 1.0*, with third-party rights preserved. The license and third-party notices accompany the source.

D Source state and reproducibility

The recorded build, dependency audit, Nanoda replay, Comparator run, and statement inspections concern the release snapshot at commit `935474b0c3678e053e1152a49ed6593bd8ad44c9` (18 September 2026) [12]. All 2 299 recorded project-source and toolchain files matched the checked snapshot when the Comparator success and statement inspections were recorded.

The line measurements below refer instead to the development snapshot of 17 September 2026, 07:34:33 UTC. The recorded verification runs do not remeasure this earlier inventory or its declaration footprints.

D.1 Source inventory

The measured development inventory contains 2 295 tracked Lean files under `Atlas/`, with 161 772 lines, and the root driver `Atlas.lean`, with 706 lines. The total is therefore **162 478 lines in 2 296 files**. This includes audit and helper modules, comments, and blank lines; it excludes untracked drafts, Mathlib, and verification-generated Lean files. The small Ree-specific modules, including finite-field support and the public package, account for 54 files and 3 470 lines, without shared generic infrastructure.

The physical file sizes are distinct from the declaration-range footprints in Table 7, which count only source lines used by the selected principal results.

D.2 Build and dependency audit

The environment is pinned to Lean `leanprover/lean4:v4.34.0-rc2`, compiler commit `6a10ac8c22beadecabdbb0919c2b50214762f91d`, and Mathlib revision `85e3a25e006c35636f0e53b0e9296caca2685bc0`. From the release root, with that toolchain installed, run

```
lake exe cache get
LEAN_NUM_THREADS=2 lake build
LEAN_NUM_THREADS=2 ruby verification/check.rb
```

The first command fetches the pinned dependency cache. The remaining commands build the configured ATLAS targets and run the release audit; they do not invoke Nanoda or Comparator. Preserve `lean-toolchain`, `lakefile.toml`, and `lake-manifest.json`. The release catalogue `docs/catalogue.json` maps claims to declarations and evidence, and `AUDIT.md` documents the build and dependency checks.

The final build/audit driver ran from 05:43:04 to 05:52:59 UTC on 18 September. This is the final driver’s elapsed interval, not the time for a source rebuild from scratch. A preceding attempt had a process exit with status 137. The affected target was then built successfully on its own, and the final full driver passed.

D.3 Replay and statement-inspection results

Nanoda. The checker revision is `4c544ed4099c8227f07d5de77ad1e69fb0740a27`. The run checked 70 891 exported declarations, exited with status zero, and finished at 06:40:26 UTC on 18 September. Checking took 31 minutes 0.82 seconds on one thread, with maximum resident memory 5 375 116 KiB (about 5.13 GiB); export took 93.66 seconds. The exported data occupied 936 181 371 bytes.

Comparator. The run used Comparator revision `2312244ac716564a61cc0bf4e107d9abf1757a61`, exporter revision `cacf989bd75f608700820f6afc595f32e7a99a4d`, and the Landrun sandbox at revision `811cfff51ceaf3d9843708aa6d22e9b84ccac8b4`. It exited with status zero after 41 minutes 35.542 seconds of service runtime. The formal-reference qualification is stated in Appendix C.2.

Statement inspections. The actual carrier applications were inspected in addition to the weaker automatic test for the presence of model declarations in type dependencies.

D.4 Per-group declaration footprints

Table 7 measures the source needed by the selected order and simplicity declarations for each completed entry. The global deduplicated ATLAS footprint is **88 856 lines**. This measures source dependencies, not proof-term size, model-token usage, or the amount of new mathematics.

Each column counts physical source lines in compiler-recorded declaration ranges reached through the selected theorem’s transitive type and proof dependencies, including interior comments and blank lines. Generated helpers use the recorded enclosing declaration’s range; a final position at column zero excludes that line. Ranges are merged by file, so column and row totals overlap. Lean core and external libraries are excluded.

Group or family	Order footprint	Simplicity footprint	Union
C_p	5	12	12
Alt_n	4	4	7
$\text{PSL}_n(q)$	168	322	379
$B_r(q)$	4 121	7 898	8 009
$C_r(q)$	708	2 203	2 203
$D_r(q)$	4 290	2 466	5 408
$G_2(q)$	1 237	3 287	3 568
${}^2G_2(q)$	1 570	2 225	2 250
M_{11}	5 479	5 866	5 868
M_{12}	4 924	5 959	5 961
M_{22}	4 183	5 348	5 351
M_{23}	4 119	4 492	4 495
M_{24}	3 477	4 582	4 585
Co_1	10 826	12 334	12 346
Co_2	10 825	14 070	14 072
Co_3	12 137	16 423	16 423
McL	13 704	16 375	16 377
HS	13 447	14 771	14 776
Suz	19 727	21 320	21 374
J_2	6 989	7 954	8 184
Fi'_{24}	31 813	33 396	33 397
Fi_{23}	31 950	34 247	34 249
Fi_{22}	31 955	34 248	34 250
Global deduplicated union			88 856

Table 7: Compiler-derived ATLAS source-line dependency footprints for 46 principal order and simplicity declarations: eight families and fifteen sporadic constructions. Shared source lines are counted once within each union; different rows need not have disjoint dependencies.

For $\text{PSL}_n(q)$, n is matrix dimension; r is rank for B, C, D . The type- A , symplectic, type- B , and G_2 simplicity roots are exact-range equivalences including their exceptional cases. The Ree root applies for $q = 3^{2m+1}$, $m \geq 1$. Isomorphic construction families have separate rows.

The 46-root footprint audit has 65 676 nodes, complete graph references, and resolved source ranges for every project dependency. The broader 911-declaration release audit is a separate verification scope.

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